



Anti-fouling System Based on PDMS and Polyhexamethylene Guanidine for the Protection of Marine Vessels

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Abstract

In this study, compositions based on PDMS and MQ resin containing the well-known biocide polyhexamethylene guanidine (PHMG) were used to produce polymer films on the way to protect the surface of the steel plates. Due to the good solubility of PHMG in water, PHMG was covalently bonded to the siloxane backbone to avoid their leaching out of the film during prolonged exposure in water. The finished biocidal coatings exhibited high homogeneity on primed steel, transparency, and high adhesion. The resulting coated steel plate samples were exposed to tropical waters, then extracted and treated with water flow with high pressure. It was shown that after water treatment, majority of the fouling was removed and the plate surface was no different from the original surface.

Keywords Polydimethylsiloxan · Polyhexamethylene guanidine · Antifouling coatings

1 Introduction

Marine fouling is a significant problem associated with the attachment of marine organisms to any type of surface that is exposed to seawater for extended periods of time [1]. The initial stage of marine fouling is the formation of a thin film of microorganisms, which occurs within a relatively brief time frame (typically weeks to months). Subsequent to the initial film's formation, larger marine organisms begin to attach to the surface due to high adhesion to the microfilm surface [2, 3]. This process continues until the entire surface is covered with mussels, corals, and other organisms.

Mankind has learned to cope with marine biofouling by using paints with high heavy metal content, which has later been shown to result in significant destruction of marine flora and fauna in harbors, as well as corals [4, 5]. Following such studies, it was decided to use heavy metal-free formulations [6–10]. In parallel, following the meeting of the International Maritime Organization's Biofouling Guidelines in 2017, a number of countries have introduced regulations on ship biofouling, resulting in a law requiring ships arriving in port to be 'hull clean' [11, 12].

Today, these types of coatings can generally be divided into three broad categories: fouling release, fouling resistance and fouling degradation [13]. One of the most popular and widely used formulations is based on polydimethylsiloxane (PDMS), which possess extremely low surface energy [14–18]. In recent years, the use of copolymers, block copolymers, and PDMS-based composites has experienced a surge in popularity, primarily due to their ability to extend the properties of silicone coatings [19–27]. Anti-fouling coatings based on pure PDMS were immediately put to practical use because they effectively self-clean from fouling as the ship moves. However, such coatings based on pure PDMS are not sufficiently effective on surfaces of non-moving objects in seawater. A promising approach is to combine the effects of fouling release and fouling resistance in a single coating. Such a coating can simultaneously

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prevent the fouling of small microorganisms through the presence of antimicrobial agents, as well as reduce adhesion to protect against the adhesion of microorganisms and larger organisms.

Despite the significant advancements in existing formulations in the domain of marine antifouling, the development of more economical, effective, and versatile antifouling coatings with high adhesion and low abrasion resistance that maintain excellent protection in both steady-state and dynamic conditions remains challenging. Many effective antifouling coatings have been developed, including those based on peptides or using advanced polymerisation techniques, but they are often complex, expensive copolymers that are difficult to obtain for rapid mass application [28–32]. This paper demonstrates that the incorporation of simple and commercially available organosilicon compounds in conjunction with the well-known and readily available biocide polyhexamethyleneguanidine (PHMG) [33–37] can yield antifouling coatings that are comparable in performance to more complex and expensive formulations. The developed coating exhibits dual functionality, combating both the fouling of microorganisms due to the biocidal properties of PHMG and reducing the adhesion of large fouling agents. Despite the prolonged duration of fouling, these fouling agents can be readily eliminated through water pressure, thereby ensuring the stability of the coating.

2 Experimental Section

2.1 Methods and Materials

All chemicals were reagent-grade and used without purification. Allyl glycidyl ether (97%, Alfa Aesar, US), 3-Glycidoxypropyltrimethoxysilane (97%, Acros Organics, Belgium), hexamethylcyclotrisiloxane (98%, Acros Organics, Belgium), polyhexamethylene guanidine (PHMG) (98%, ECOS-1, Russia), 3-aminopropyltriethoxysilane (APTES) (96%, ABCR, Germany), α,ω -hydroxypolydimethylsiloxane «SKTN-A» (SAZI, Russia), Platinum(0)–1,3-divinyl-1,1,3,3-tetramethyldisiloxane complex solution, in xylene, Pt~2% (Karstedt's catalyst) (Acros Organics, Belgium), Epoxy primer (Reoflex, Russia), methyl tertbutyl ether (MTBE), (EKOS-1, Russia).

Hydride terminated polydimethylsiloxane with a molecular weight of 1800 was obtained by [38]. We used commercially available PDMS (SKTN A) premodified with 3-aminopropyltriethoxysilane (PDMS-t-APTES). PDMS-t-APTES, based on “SKTN A”, had $M_w=31,500$, $M_w/M_n=1.7$ containing 0.14 wt% of amino groups. The method for its preparation was described previously [39]. The cross-linking agent was the MQ copolymer $[\text{SiO}_2]_{1.32}[\text{Si}(\text{OH})\text{O}_1$

$_{5}]_{0.68}[\text{O}_{0.5}\text{Si}(\text{CH}_3)_3]$ with the ratio $M : Q = 1 : 2$, containing 5.8 wt% OH groups, $M_w = 14,200$ Da, $M_w/M_n = 1.3$. The procedure for obtaining MQ copolymer was described previously [40].

2.2 Equipment and Measurements

^1H spectra were recorded on Bruker Avance AV-300 spectrometer (300 MHz for ^1H).

Gel permeation chromatography (GPC) analysis was performed on a Shimadzu LC-10 A series chromatograph (Japan) equipped with a RID-10 A refractometer and SPD-M10A diode matrix detectors. Analytical separation was performed using a 7.8 mm $\times$ 300 mm Phenomenex column (USA) filled with Phenogel sorbent with a pore size of 15–500 Å. THF was used as the eluent.

Pneumatic sprayer “Caliber” brand KRP-1.0/0.12 VB-profi with a working chamber volume of 125 ml on pre-degreased duralumin plates measuring 80 $\times$ 50 $\times$ 3 mm or 80 $\times$ 15 $\times$ 3 mm.

A scanning microscope JSM-6000PLUS (JEOL, Japan) was used to study the surface morphology of the samples.

The elemental composition of each surface was determined using the EX-230**BU with an integrated energy dispersive analysis system.

2.3 Synthesis of PDMS-Gly

Epoxy-containing PDMS (PDMS-Gly) was obtained through a hydrosilylation reaction between hydride terminated polydimethylsiloxane with a molecular weight of 1800 (11 g, 6.1 mmol) and allyl glycidyl ether (1.4 g, 12.2 mmol). The reaction mixture was stirred at 80 °C for 12 h in 13 ml dioxane in the presence of 7 μl Karstedt's catalyst [41]. The isolation of the target product was achieved through (SiO_2 ; hexane-dichloromethane=1:1) at a yield of 91% (10.2 g, 6.1 mmol).

^1H NMR (CDCl_3): δ 3.72–3.66 (m, 1H, -O-CH- CH_2), 3.51–3.34 (m, 3 H, - CH_2 -CH, - CH_2 -O), 3.17–3.11 (m, 1H, - CH_2 -O), 2.81–2.77 (m, 1H, O- CH_2 -CH), 2.62–2.59 (m, 1H, O- CH_2 -CH), 1.67–1.55 (m, 2 H, - CH_2 - CH_2 -), 0.56–0.49 (m, 2 H, -Si- CH_2 -), 0.29–0.03 (m, 70 H, -Si- CH_3).

2.4 Synthesis of Mixture Containing PHMG

The mixture (3-Glycidoxypropyl)trimethoxysilane (1.6 g, 6.7 mmol) and hexamethylcyclotrisiloxane (4.2 g, 20 mmol) was dissolved in 10 mL of ethanol. Then PHMG (2 g) was added and the reaction solution was stirred at 80 °C for 6 h. The solvent was removed under reduced pressure (1 Torr). The mixture was a cloudy, viscous, milky liquid.

2.5 Synthesis of Antifouling Surface

To prepare composite films or coatings, **PDMS-t-APTES** and **MQ copolymer**, **PDMS-Gly** were dissolved in methyl tert-butyl ether (MTBE). The final concentration of the of the reactants in the solution is 20 wt%. The ratio of reagents in this step **PDMS-t-APTES**: **MQ** : **PDMS-Gly**=2 : 1 : 0.15. In the final step, a **mixture containing PHMG** was added to the reaction mixture so that the concentration of PHMG was 6–20 wt% relative to the mixture of previously added reagents. The mixture was stirred at 20 °C on a magnetic stirrer until a homogeneous solution was obtained. The mixture was applied to pre-primed metal substrates by airless spraying and allowed to cure for 16 h at room temperature. The thickness of the polymer coatings was $150 \pm 10 \mu\text{m}$. The comparison sample was primed aluminum plates without coating.

3 Results and Discussion

The main problem with incorporating PHMG into a siloxane matrix is their incompatibility. Siloxane polymers are more hydrophobic in nature and dissolve poorly in polar solvents. In contrast, PHMG is more hydrophilic and dissolves well in protic polar solvents such as water and alcohols. Thus, when attempting to attach PHMG to PDMS, phase separation of two phases is often observed, which is unacceptable for the development of biocidal coatings. Consequently, the primary objective of this study was to identify optimal and commercially available methods to obtain a PDMS matrix with PHMG beams grafted to it.

3.1 Synthesis of Antifouling Agent Copolymer and Crosslinker

The main challenge in the realization of our proposed scheme of obtaining the coating was poor compatibility of PHMG and polysiloxane components. Therefore, at the first stage of creation of antimicrobial-antifouling agent based on PHMG, we developed a one-step scheme of its modification with short polydimethylsiloxane fragments and introduction of triethoxysilyl fragments into its structure (Fig. 1a). The synthetic scheme shows the proposed structure of the resulting product. The PHMG addition product was prepared by refluxing (3-glycidoxypropyl)trimethoxysilane, PHMG and hexamethylcyclotrisiloxane (D3) in ethanol for 6 h until a clear homogeneous solution was obtained. Then 50% of the solvent was evaporated. In this case, PHMG acts as both a base and a weak nucleophile. It is known that the secondary amino group of PHMG can react as a nucleophile [42], in this case by opening the oxirane cycle of

Gly-TES (Fig. 2). Furthermore, as previously described and proposed, organic guanidine derivatives, as strong nitrogen-containing bases, catalyze the polymerization (ROP) of D3, in the presence of water, silanols or alcohols as initiators, leading to the formation of PDMS [43, 44]. Therefore, we presume that in our proposed single-reactor synthesis, there is a simultaneous addition of Gly-TES to PHMG as well as the opening of cyclosiloxane D3 in the presence of ethanol. Furthermore, due to the presence of the same PHMG, heterofunctional condensation can occur between the resulting silanol groups of the oligomeric polydimethylsiloxane and the Gly-TES embedded in the PHMG structure, leading to the formation of the final modifier, as shown in the scheme. Figure 1(c) presents the GPC curve of the formed product, and its molecular weight was determined to be 1600 Da according to the GPC data.

Also, according to a well-established scheme, PDMS-Gly was prepared as a crosslinking agent by hydride-terminated polydimethylsiloxane and allyl glycidyl ether hydrosilylation reaction (Fig. 1b).

The completeness of the hydrosilylation and the structure of PDMS-Gly were confirmed by ^1H NMR methods (Fig. 2). The completeness of hydrosilylation was confirmed by the appearance of signals of methyl protons at silicon at $\delta=0.53$ ppm and the disappearance of signals corresponding to the proton of the hydridesilyl group at $\delta=4.60$ ppm. The product obtained was also analyzed by GPC. The product was characterized by a monomodal distribution with $M_w/M_n=2.5$, $M_n=1700$ and $M_w=3800$.

3.2 Fabrication of Antifouling Coatings

In order to obtain antifouling compositions that can perform in marine waters, it was necessary to select the composition in such a way that the resulting coating is not brittle, has suitable physico-mechanical properties and is resistant to the action of seawater. For this reason, formulations with a complex composition of components were chosen.

For the siloxane polymer matrix, we chose a composition consisting of functionalized polydimethylsiloxane with terminal epoxy and amine groups and low molecular weight MQ resins containing residual silanol groups as fillers. To obtain crosslinked systems based on these two components, the initial silanol-terminated polydimethylsiloxane was functionalized with aminopropyltriethoxysilane (APTES), resulting in the formation of a soluble product with aminopropyl- and ethoxysilyl-terminated groups (PDMS-t-APTES) (Fig. 3). “SKTN A” is a widely available commercial reagent, and including PDMS with such a molecular weight and MQ copolymer within our specified parameters provides the best mechanical characteristics. MQ is an efficient molecular filler that dissolves perfectly in

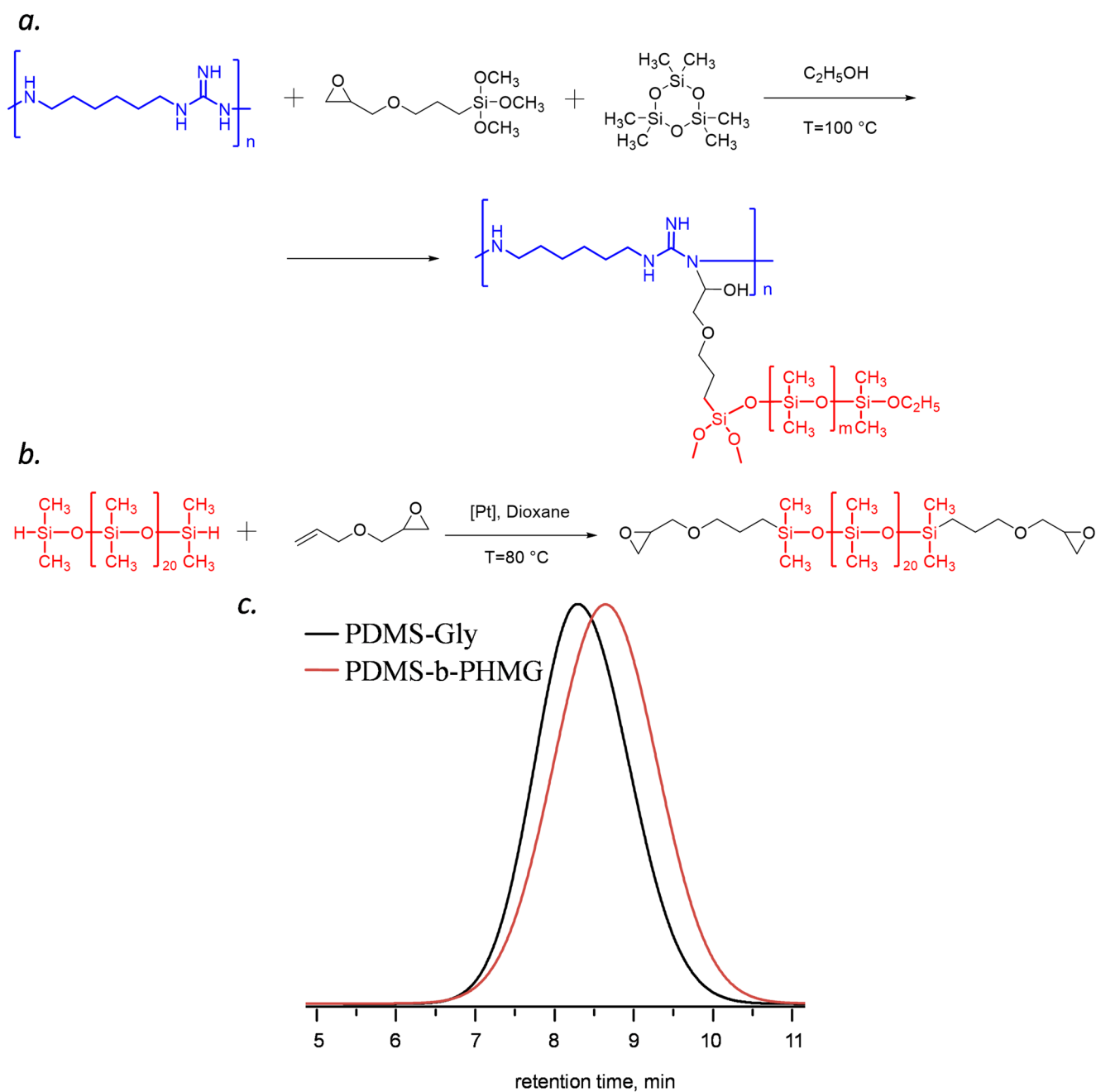


Fig. 1 Synthesis scheme of PHMG: PDMS-b-PHMG (a), synthesis scheme of PDMS-Gly (b) and GPC of PDMS-Gly and its adduct with PHMG: PDMS-b-PHMG (c)

PDMS and is suitable for adjusting the mechanical properties of PDMS-based compositions [39, 45, 46].

The following components were selected to obtain composites with a mass content of 6, 10, and 20% PHMG: PDMS-Gly, PDMS-t-APTES, and MQ, as well as a mixture of PDMS-b-PHMG functionalized by methoxy groups and dimethylsiloxane oligomers (see Fig. 3). Crosslinking occurs due to the interaction of various functional groups contained in almost all components, including the filler. The

epoxy-containing PDMS and the mixture containing PHMG were then utilized as antifouling modifiers.

The data concerning the solutions prepared with varying PHMG content by weight of all starting substances are presented in Table 1. The Biocide component with PHMG PDMS-b-PHMG was incorporated into three soluble matrix coatings. The first formulation contained the Biocidal component with 6 mol% in terms of PHMG, the second formulation contained 10 mol%, and the third formulation contained 20 mol% PDMS-b-PHMG.

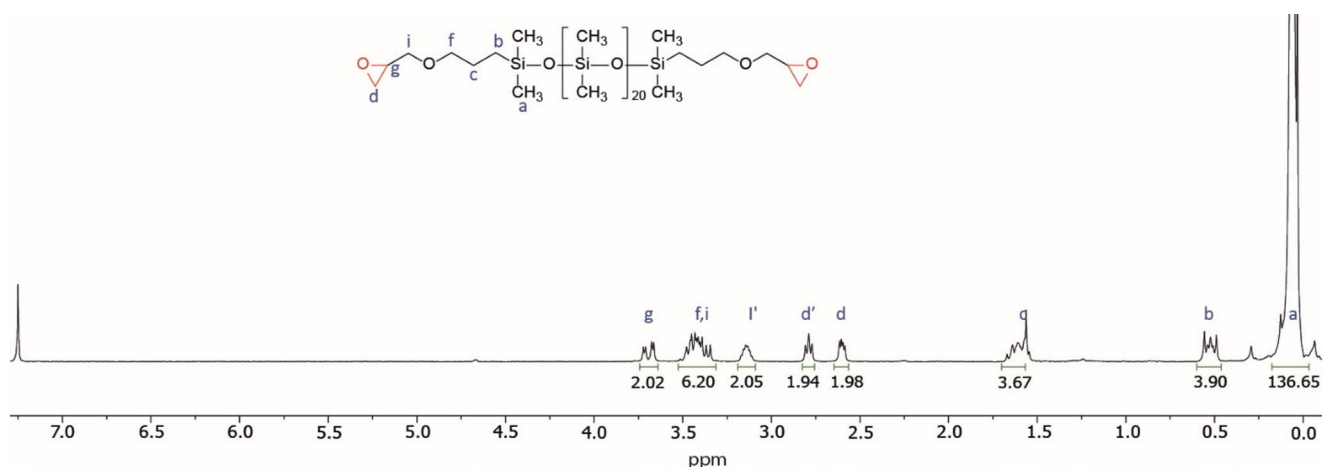


Fig. 2 ^1H NMR spectrum of the obtained PDMS-Gly

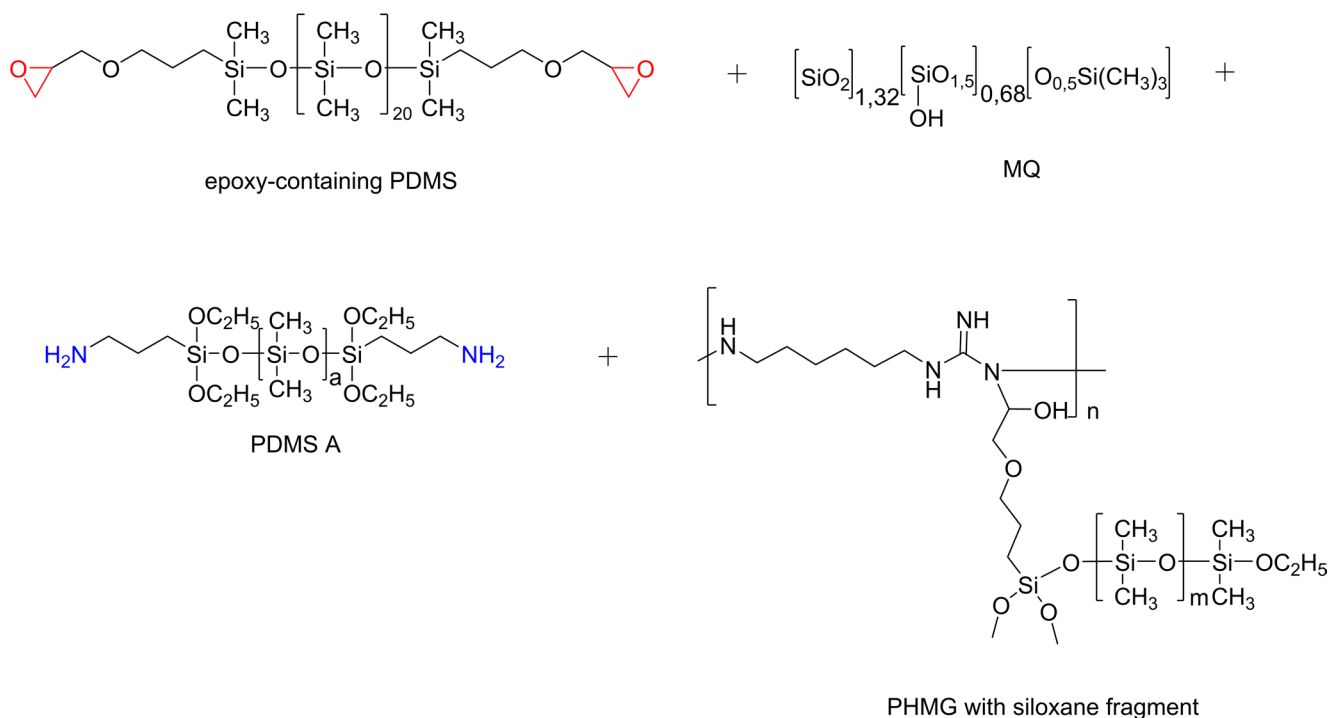


Fig. 3 Synthesis scheme of the mixture containing PHMG

Table 1 Composition of PDMS-b-PHMG based antifouling coatings

#	Compound	PHMG content, %	Solvent	№ Plates
1	PDMS-t-APTES+MQ+PDMS-Gly+PDMS-b-PHMG	6	MTBE	1
2	PDMS-t-APTES+MQ+PDMS-Gly+PDMS-b-PHMG	10	MTBE	2
3	PDMS-t-APTES+MQ+PDMS-Gly+PDMS-b-PHMG	20	MTBE	4
0	Only primed surface	-	-	0

3.3 Preparation of Testing Samples

Basic surface preparation was performed - smoothing, cleaning and priming. The surfaces were cleaned with compressed air or brushes, degreased with acetone and dried. The film of the primer - modifier was visually checked for the absence of wrinkles and cracks. The adhesion of the film was checked after 1 day using the cross-cut test.

The composition was applied using the technology of airless application of paint and protective materials. When applied to a dry surface with a humidity level ranging from 2 to 3.8%, the material exhibited optimal adhesion properties.

This method of applying the solution of compositions on steel plates and subsequent drying has been demonstrated to result in the formation of homogeneous, glossy coatings.

The composition was thoroughly mixed for at least 10–15 min at room temperature. The composition was applied evenly on the treated surface with a spray gun under pressure of 3–5 atm. The solution was applied in 3 coats, with 15 min between coats until the tack disappeared. The thickness of one coat was 50 μm . The thickness of the final surface was 150 μm . Drying time was 16 h at room temperature. As we can see in Fig. 4a, this application method was very effective: the coatings were quite uniform and homogeneous, except for artifacts from the primer.

3.4 Antifouling Tests

At the present of antifouling coating development, there are two most widely used mechanisms to prevent surface fouling by marine organisms - antifouling biocide coatings and fouling release coatings. Since the scientific community is trying to find affordable and biologically harmless coatings, in this work we have chosen compositions without the addition of biocides harmful to marine organisms. Thus, a self-polishing mechanism of antifouling was chosen to clean the surface from microorganisms. This mechanism of antifouling protection is based on the low surface energy of the protective coating, as a result of which deposited micro- and macrofouling organisms are not firmly attached to the surface and are mechanically removed as the speed of the ship's movement increases [47]. As a simulation of increased vessel speed, in this paper, as in other similar work, we used a strong water pressure as a means of getting rid of low-adhesive organisms.

To evaluate the Biofouling rate, samples of antifouling coatings applied to pre-finished and primed steel plates of $2 \times 100 \times 100$ mm were collected in cassettes and placed in water at a depth of 2 m (Fig. 4a, b). The coating applied to the protected surface was tested for 9 months under tropical marine conditions (Dam Bai Climate Station, Nha Trang Bay, Che Island) and showed resistance to fouling. Tropical waters had the following parameters for 9 months: temperature from 25.7 to 31.1 °C; oxygen from 5.8 to 8.4 mg/l, salts from 30.7 to 35.5‰.

The first time the plate cassettes were removed from the water after three days of testing. After only 3 days, the untreated back side of the plates was already biocorroded (Fig. 4b), while the treated side did not change at all (Fig. 4c).

After 9 months of full-scale climatic testing, the sample cassettes were removed from the water. As can be seen, the sample cassette had been exposed significant marine fouling (Fig. 5a). A pressure washer was used to treat the surface of

the samples from the biofouling. Total (China) equipment with a maximum water pressure of 24.8 bar was used and each surface was treated for 10 min with a gradual increasing pressure. Photos of the cassette with the most biofouling resistant samples before and after treatment are shown in Fig. 5b.

As can be seen from the photos above, panels 1, 2, and 4 are the most resistant to biofouling. The biofilm has almost completely washed away from their surfaces, except for the attachment sites. For clarity, Figs. 5c–g show a close-up of plate 1 before the water pressure treatment (Fig. 5a) and after the treatment at different water jet pressures. As can be seen, the coating in Fig. 5g has almost regained its original appearance. The fouling on the edges of the plate remained undeveloped, apparently due to the increased adhesion of the microorganisms to the steel.

It is worth noting that further increasing of the water pressure lead to destruction and delamination of the primer's layer with exposure of the plate surface. This indicates a gradual decrease in the adhesion properties of the epoxy primer to steel during prolonged exposure to salty seawater.

To determine the extent of damage to the protective film, the surface microstructure of these samples was examined.

3.5 Analysis of the Surface of Steel Samples After Water Treatment

The analysis of the surface and biocorrosion, carried out using scanning electron microscopy (SEM) to study the microstructure of the biofilm formed on the surface of the steel sample, showed its high degree of resistance to fouling. Figure 6a shows microphotographs of the surface of sample 1.

The surface is mostly smooth and homogeneous, with some areas that are not completely cleaned. Based on additional Energy Dispersive Spectroscopy (EDS) analysis with determination of the element composition of the surface, it was determined that these are calcium carbonate deposits, probably left by mollusks, sponges or coral polyps (Fig. 6b, d).

The same picture is observed for sample 2 (Fig. 7). There are no chips, cracks on the surface of the coatings and, as in the case of sample 1, there are also organogenic residues, but their size does not exceed 12 μm . Sample 3 was more severely damaged. It is likely that the expanded on the surface organisms were able to penetrate deep into the coating, but not deep enough to withstand the pressurized water scrubbing, hence the scratches on the surface of the sample.

From the microscopy data, it can be seen that firstly, an antifouling coating is present in all the samples examined, as demonstrated by the EDS method. Secondly, mechanical damage due to strong adhesion of microorganisms is

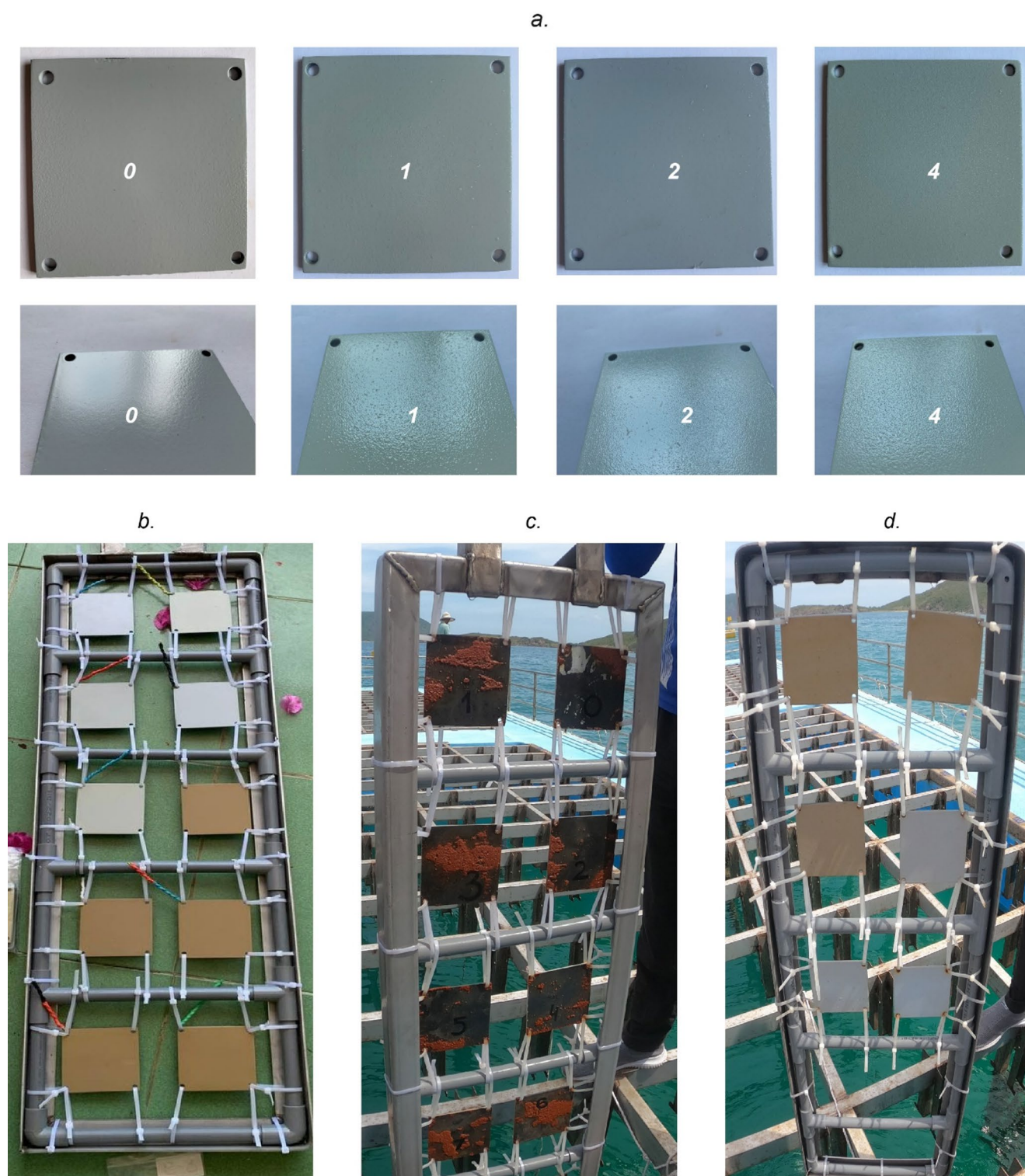


Fig. 4 Prepared steel comparison plates (0) and with applied antifouling coatings – 1, 2, 4 (**a**); initial cassette assembled with the tested samples (**b**); back uncoated side of the plates after exposure for 3 days (**c**) and working surface of the plates (**d**)



Fig. 5 Surface photos of samples after 9 months of exposure to static water: **a** before pressurized water treatment, **b** after pressurized water treatment; close-up photos of sample 1 before pressurized water treatment (**c**) and during treatment (**d–g**)

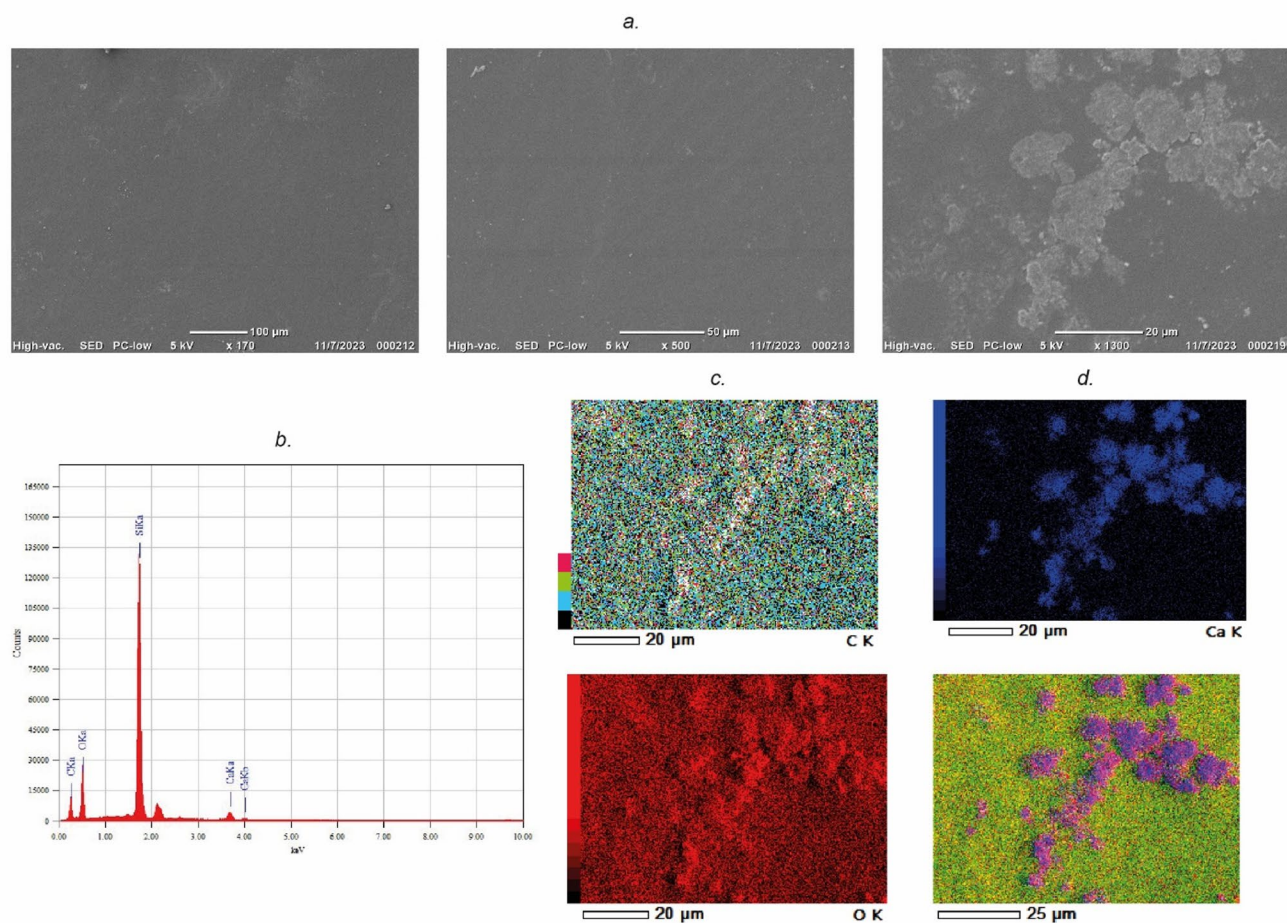


Fig. 6 SEM images of the surface of sample 1 after 9 months exposure in static water (a) EDS (b) and microphotographs (c, d) of the surface of sample 1

visualized on the microphotographs, but the geometric dimensions of the damaged areas are tens of microns.

4 Conclusion

In the present work, compositions based on siloxanes with PHMG were obtained. By optimizing the cross-linking components and other auxiliary compounds, compositions were obtained that formed strong transparent films on the surface of primed steel. Compositions with 6, 10 and 20% Biocidal component in terms of PHMG were applied to primed steel plates and the samples were exposed to tropical oceanic waters of Vietnam for 9 months. The marine fouling that appeared on the plate surface was then treated with water pressure to simulate the self-polishing effect, which in the case of antifouling coating samples with PHMG resulted in almost complete exfoliation of the marine organisms. Analysis of the panel surface by SEM with EDS revealed

areas of micron-sized coating degradation associated with mechanical damage to the coating by microorganisms. Additional studies of the elemental composition by EDS method confirmed the presence of a film of antifouling composition on the plate surface.

Thus, in this study, effective and commercially available silicone antifouling coatings based on PHMG without releasing any harmful for marine organisms agents were developed.

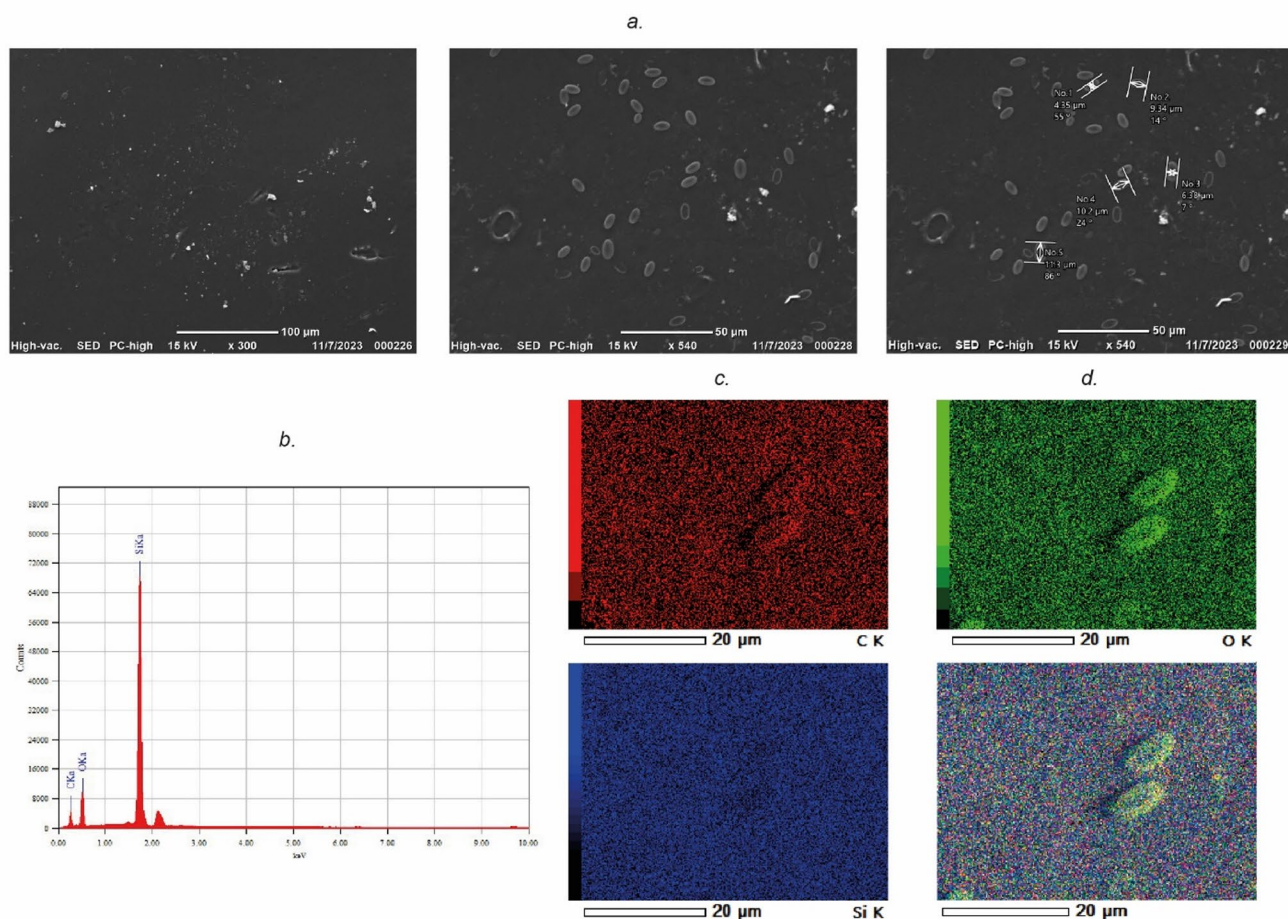


Fig. 7 SEM images of the surface of sample 2 after 9 months exposure in static water (a) EDS (b) and micrographs (c, d) of the surface of sample 2

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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